

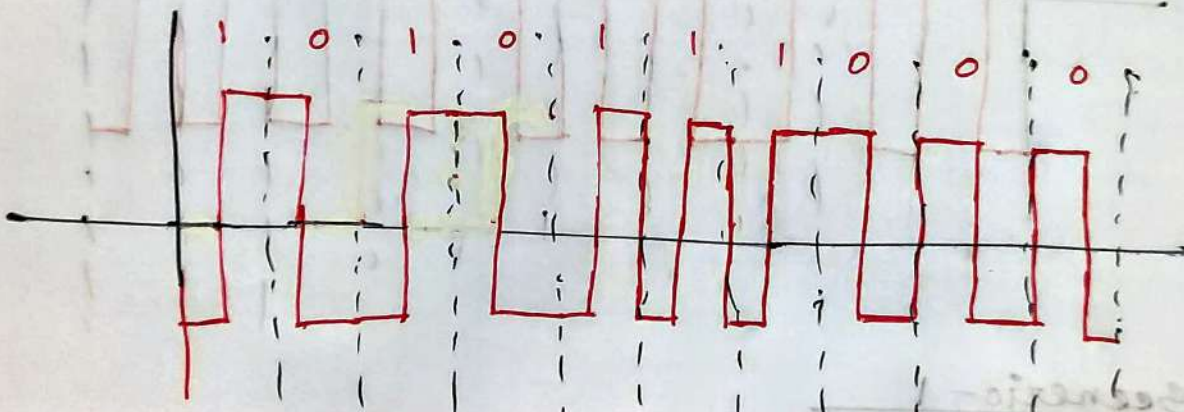
Q₁
Signal A → 1010111000

Signal B → 1110111011

AND 1010111000

No consecutive zero problem · use MLT-3 / Manchester / Differential Manchester

using Manchester



Q₂

3 bit → 4 bit

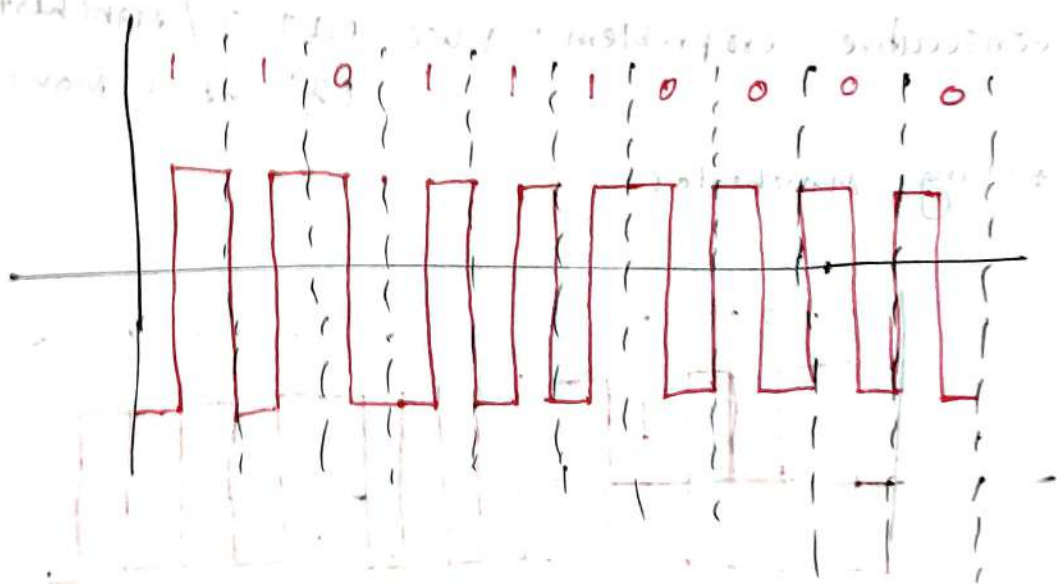
↓
 $2^3 = 8$ combination

3 bit	4 bit
111	0010
110	0011
101	0100
100	0101
011	0110
010	1001
001	1010
000	1011

Q3

① 1101110000

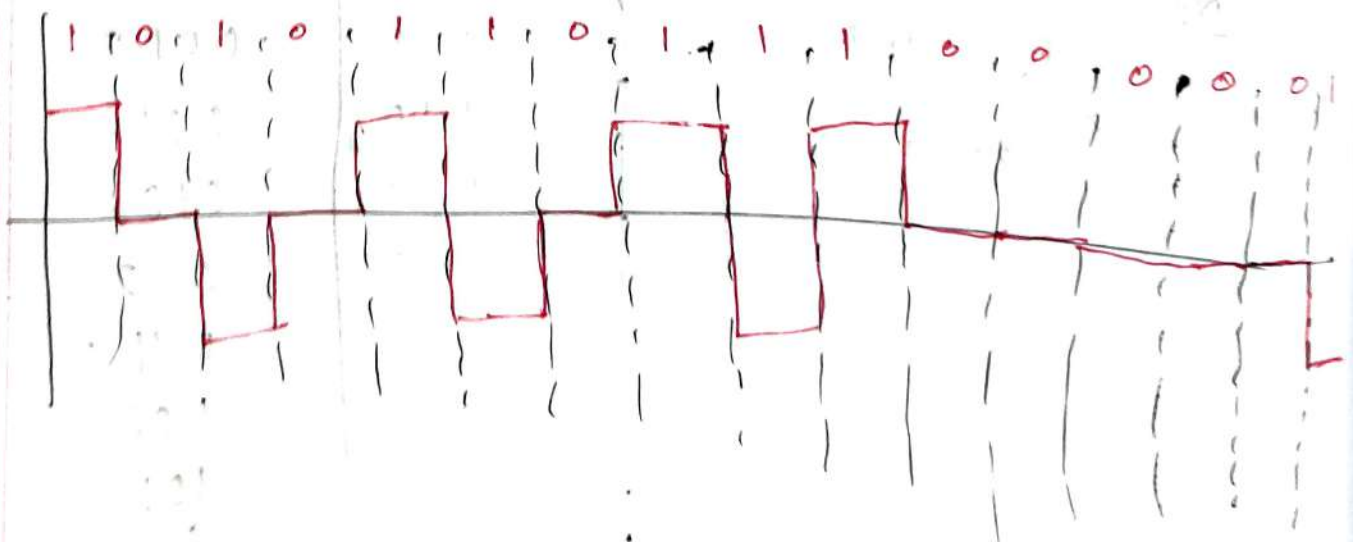
②



Q4

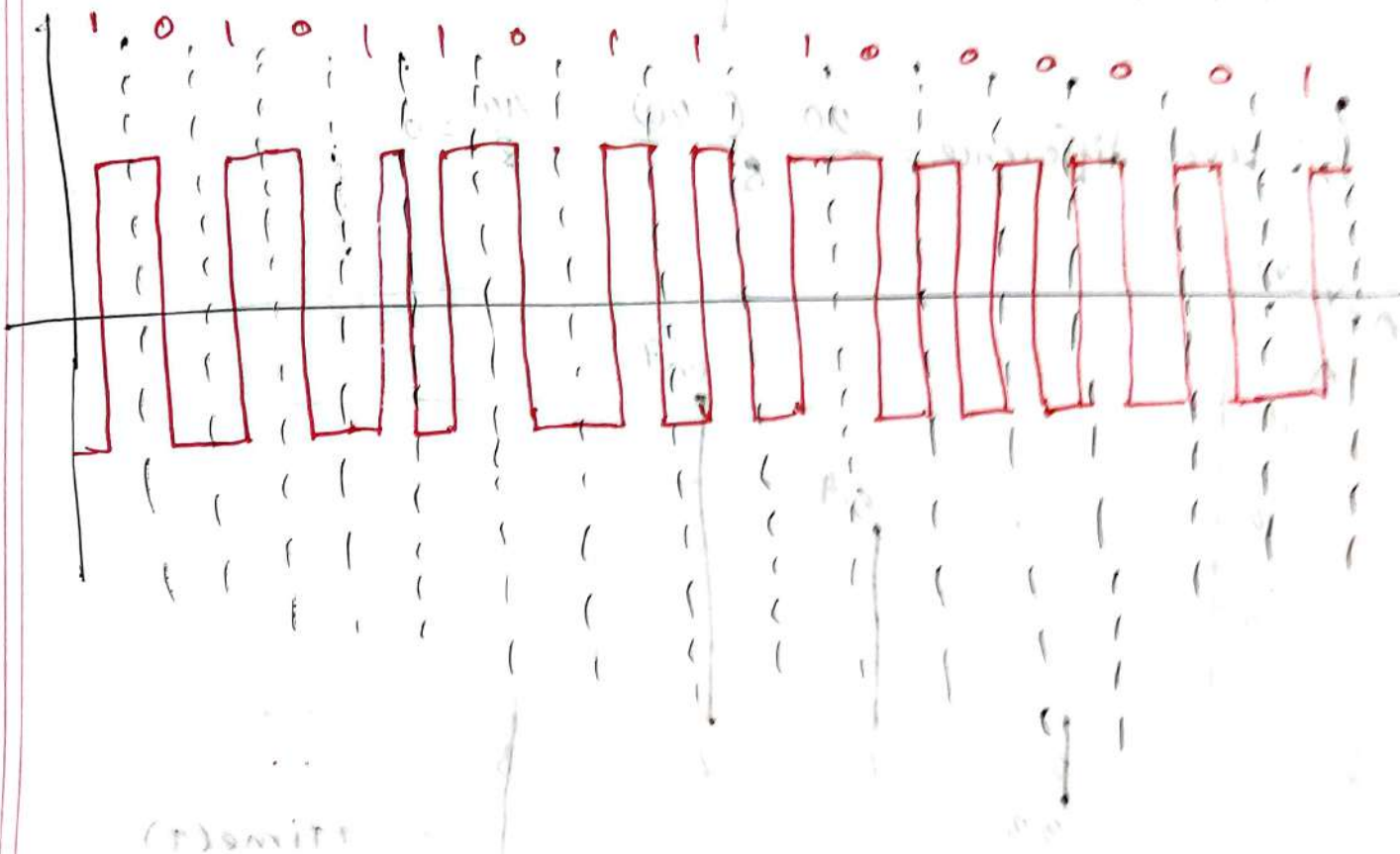
Scenario-1

AMI $\rightarrow$ no Base line Wandering, low broadband



Scenario-2

Use Manchester / Differential Manchester



(1) transmit

0.07

10.2	11.2	12.2	13.2
2.2	2.2	2.2	2.2
11.2	12.2	13.2	14.2
15.2	16.2	17.2	18.2
19.2	20.2	21.2	22.2
23.2	24.2	25.2	26.2
27.2	28.2	29.2	30.2
31.2	32.2	33.2	34.2
35.2	36.2	37.2	38.2
39.2	40.2	41.2	42.2
43.2	44.2	45.2	46.2
47.2	48.2	49.2	50.2

Q5

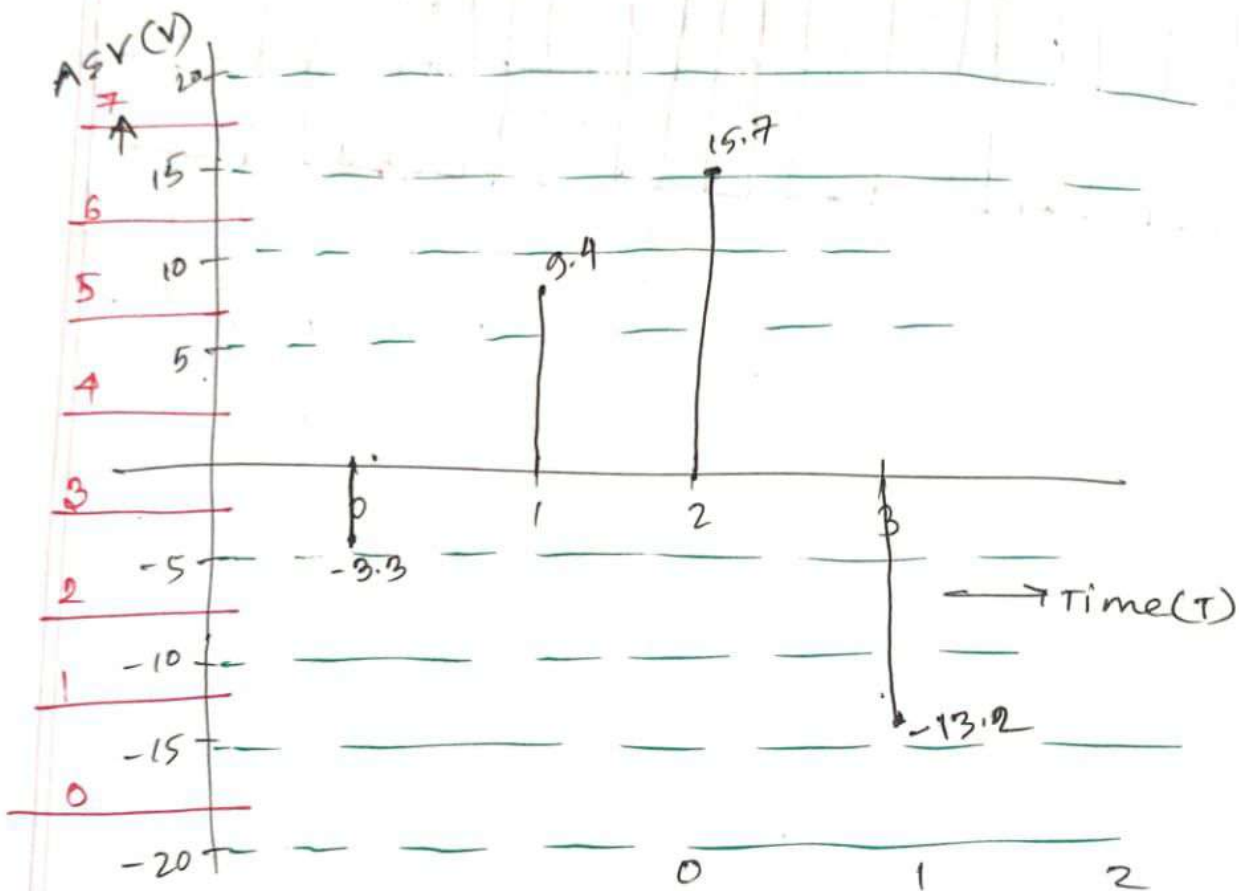
Max Amplitude = 20 V

Min Amplitude = -20 V

of bits = 3

$\therefore$ level = $2^3 = 8$

$$\therefore \text{Level Difference} = \frac{20 - (-20)}{8} = \frac{40}{8} = 5$$



Normalized PMA value -0.66 1.88 3.14 -2.64

✓ Normalized quantized value -0.5 1.5 3.5 -2.5

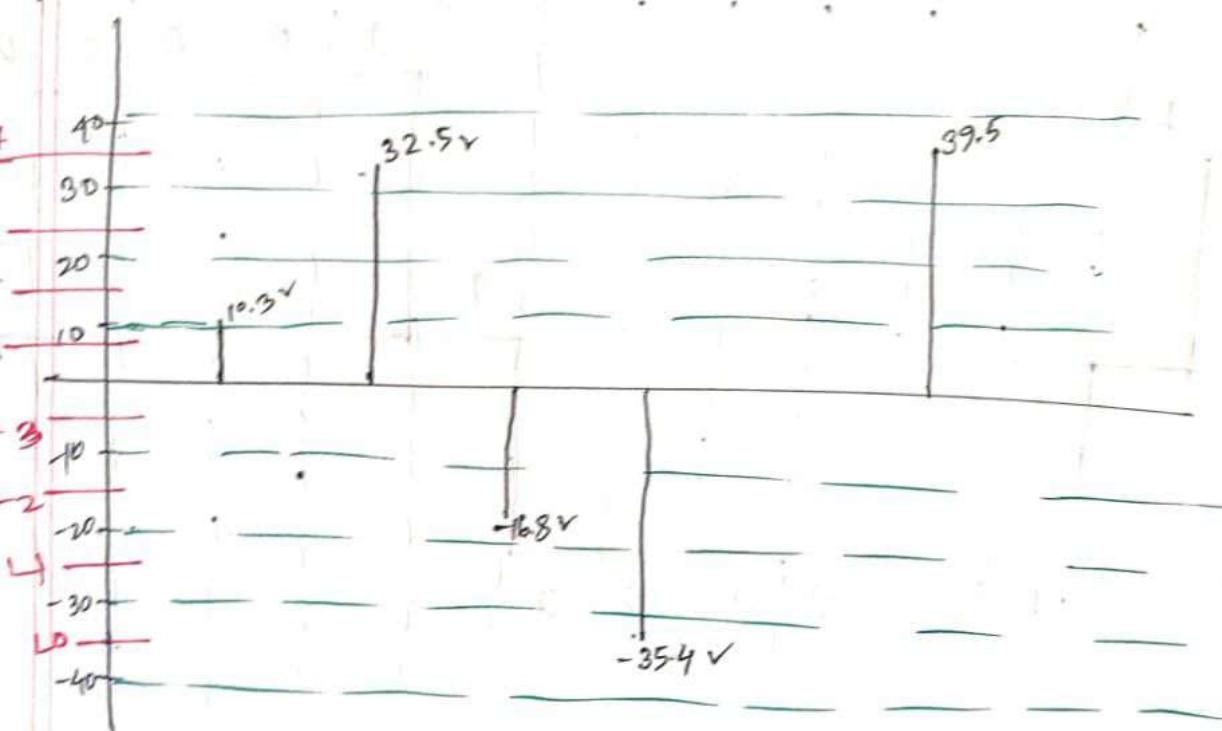
Error 0.16 -0.38 0.36 0.14

Quantized code 3 5 7 1

✓ Encoded value 011 ~~010~~ 101 111 001

Q6

Level = $2^3 = 8$ | Level Difference = $\frac{80}{8} = 10$



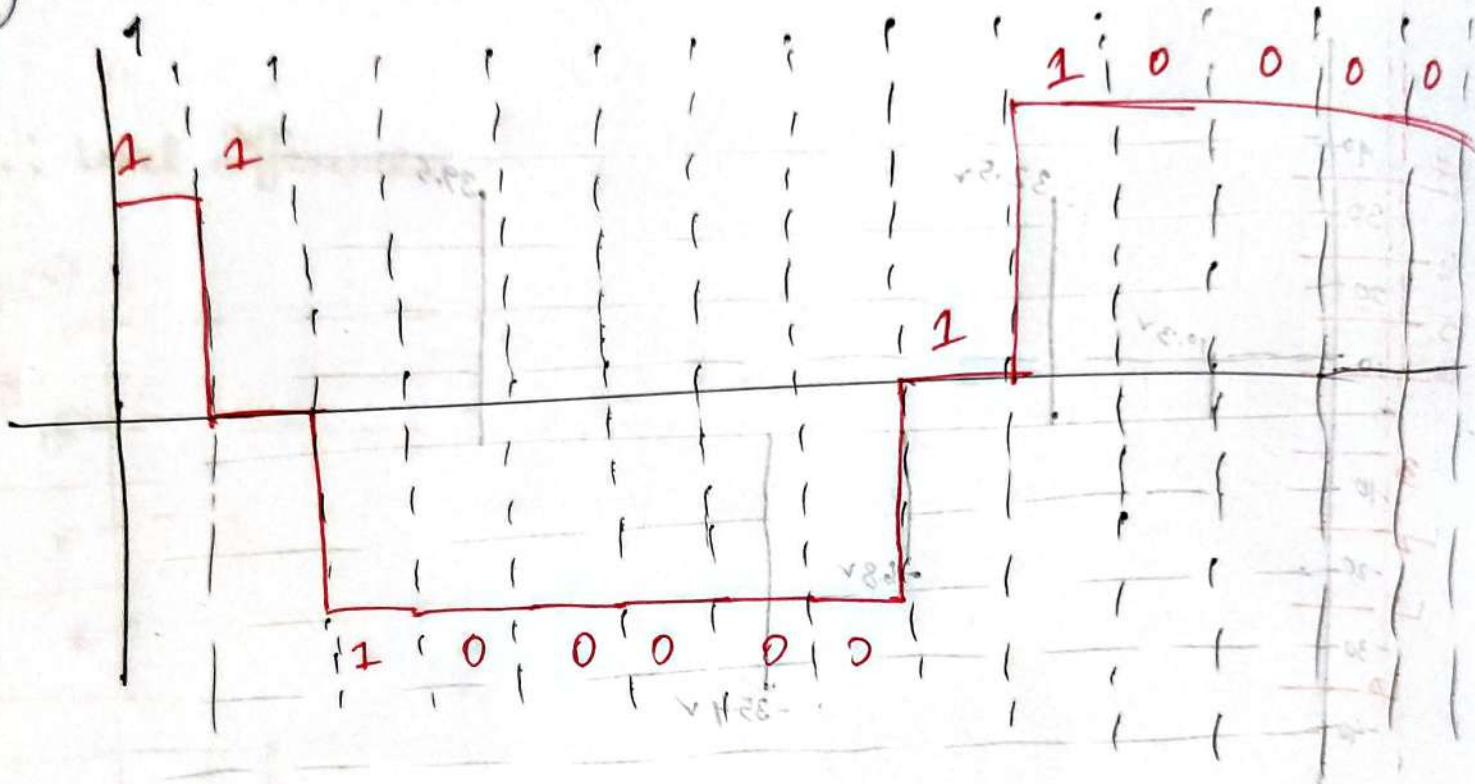
	1	2	3	4	5
Normalized DAW value	1.2875	4.0625	-2.1	-4.425	4.9375
Normalized quantized value	1.5	3.5	-1.5	-3.5	3.5
Quantization code	5	7	2	0	7
Encoded value	101	111	010	000	111

Q-8

MLT-3

Level = 8, Level Difference = 8

(i)



(ii)

Baseline wandering → consecutive 0.

DC component → No

(iii)

HDB3

EVEN

8 0 0 0 V

1 1 1 0 0 0 0 0 1 1 0 0 0 0

